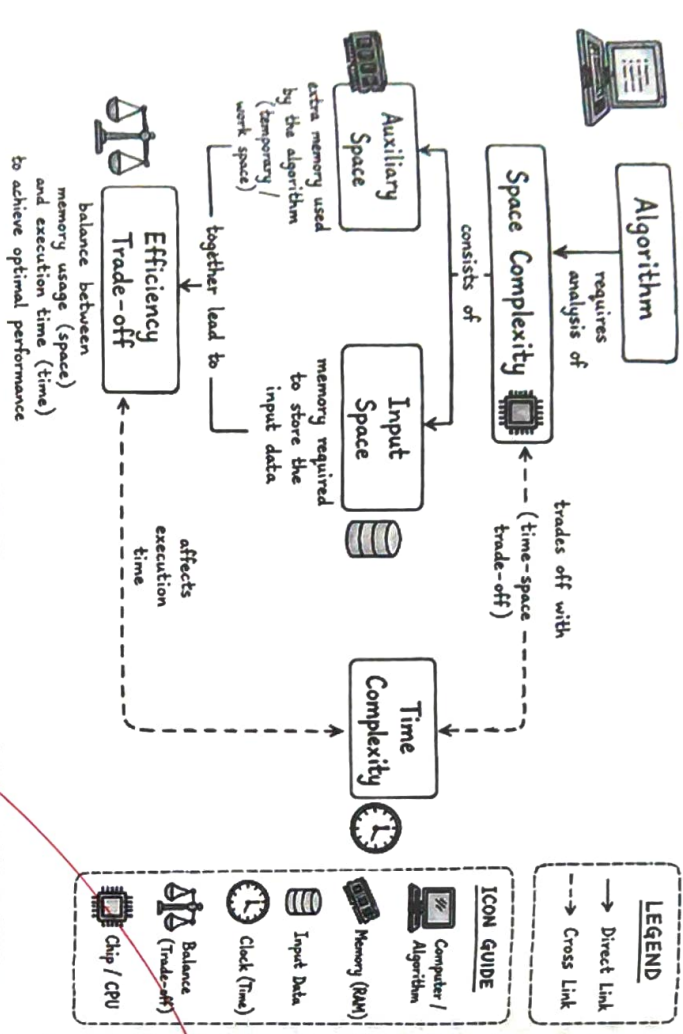


Q8

CONCEPT MAP : Algorithm → Space Complexity → { Auxiliary Space, Input Space } → Efficiency Trade-off
 Cross-link : Space Complexity ↔ Time Complexity (Time-Space Trade-off)

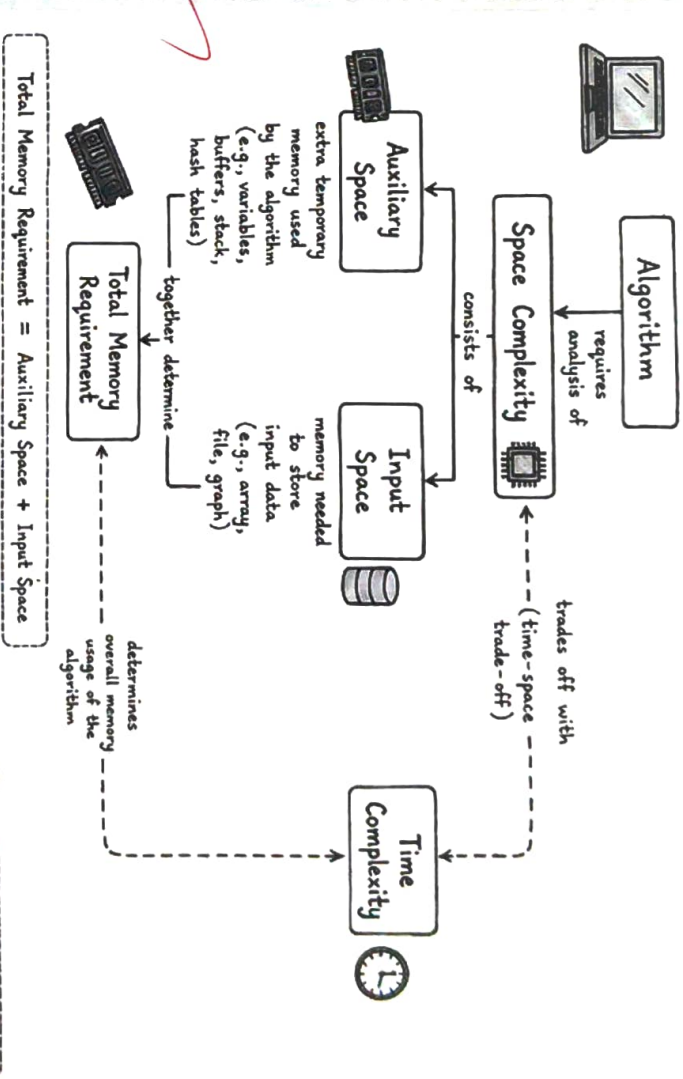


- 1. CONCEPT IDENTIFICATION (LIST)**
 - Algorithm → Step-by-step procedure to solve a problem.
 - Space Complexity → Amount of memory used by an algorithm.
 - Auxiliary Space → Extra memory used apart from input.
 - Input Space → Memory required to store the input.
 - Efficiency Trade-off → Balance between space (memory) and time (execution).
 - Time Complexity → Amount of time an algorithm takes to run.
- 2. EXPLANATION OF LINKAGES**
 - Algorithm → Space Complexity : Every algorithm requires memory to run.
 - Space Complexity → Auxiliary Space : Part of memory used for computation.
 - Space Complexity → Input Space : Part of memory used to store input.
 - {Auxiliary, Input} → Efficiency Trade-off : More space can reduce time or improve performance, while less space may increase time.
 - Space Complexity ↔ Time Complexity : The time-space trade-off shows that improving one often affects the other.
- 3. WHEN TRAVERSING TOWARD ACCEPTABLE HIGHER SPACE USAGE IS**
 - When time efficiency is critical (e.g., real-time systems).
 - When memory is abundant but time is limited.
 - In problems requiring fast queries (e.g., hashing, memorization).
 - When it improves code simplicity and maintainability.
 - In applications like databases, AI models, caching systems.

4. INTERPRETATION - SUPPORT FOR MEMORY EFFICIENCY
 These maps help designers reason about how memory is used in an algorithm. They allow evaluation of whether extra memory (auxiliary space) is justified by the performance benefits (reduced time, better scalability). This leads to memory-efficient and balanced algorithm design.

Q23

CONCEPT MAP (EXTENDED) : Algorithm → Space Complexity → Total Memory Requirement
 Auxiliary Space and Input Space → Total Memory Requirement
 Cross-link : Space Complexity ↔ Time Complexity (Time-Space Trade-off)



- 1. CONCEPT IDENTIFICATION (LIST)**
 - Algorithm → Procedure to solve a problem.
 - Space Complexity → Total memory used by the algorithm.
 - Auxiliary Space → Extra temporary memory used by algorithm.
 - Input Space → Memory needed to store input data.
 - Total Memory Requirement → Sum of auxiliary space and input space.
 - Time Complexity → Time taken for execution.
- 2. EXPLANATION OF LINKAGES**
 - Algorithm → Space Complexity : Every algorithm uses memory.
 - Space Complexity → Auxiliary Space : Used for computation.
 - Space Complexity → Input Space : Used for storing input.
 - {Auxiliary, Input} → Total Memory Requirement : Both together determine total memory usage.
 - Space Complexity ↔ Time Complexity : Time-space trade-off exists.
- 3. WHY HIGHER AUXILIARY SPACE MAY BE ACCEPTED**
 - To achieve faster execution (e.g., memoization, caching).
 - To handle large inputs efficiently (e.g., indexing).
 - To simplify algorithm design and reduce time complexity.
 - In performance-critical applications where time is more valuable than memory.

5. INTEGRATION OF LITERATURE
 According to Cormen, Leiserson, Rivest & Stein in "Introduction to Algorithms" (3rd ed.), both time and space complexity are fundamental measures of an algorithm's efficiency. Ellis Horowitz, Sartaj Sahni & Rajasekaran in "Fundamentals of Data Structures in C" also state that many efficient algorithms (e.g., dynamic programming, hashing, caching) use additional memory to achieve faster execution, illustrating the time-space trade-off principle.

REFERENCES

- Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). Introduction to Algorithms (3rd ed.). MIT Press.
- Ellis Horowitz, Sartaj Sahni, & Rajasekaran. Fundamentals of Data Structures in C. Universities Press.
- Donald E. Knuth. The Art of Computer Programming. Addison-Wesley.



CONCLUSION : Space and time are interdependent in algorithm design. By analyzing their relationship through these concept maps, we can choose the best balance between memory usage and execution time based on the problem requirements and system constraints.

